

 सत्यमेव जयते	भारत सरकार दूरसंचार विभाग दूरसंचार अभियांत्रिकी केंद्र Government of India Department of Telecommunications Telecommunication Engineering Centre K.L. Bhawan, Janpath, New Delhi- 110001	 ISO 9001:2015
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No. 26-1/2022-FA/TEC

Date: 11.06.2024

Subject: Minutes of Tenth (10th) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 10.06.2024.

The Tenth (10th) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies, and Infrastructures for Transport, Access & home*”, reconstituted for the Study Period 2022-24, was held on 10.06.2024 under the chairmanship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. The list of participants is placed at Annexure A.

2. At the outset, Sh. Rajesh Tripathi, Director (FA) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the agenda and further proceedings of the meeting.

3. Sh. Rakesh Desai, Chair/ NWG-15 & DDG (FA) TEC, greeted all the members and welcomed all participants especially new members from academia of IIT-Kharagpur. He highlighted that the next (4th) meeting of ITU-T SG15 is scheduled at Montreal, Canada from **1-12 July 2024**. He pointed out that the last date to upload contributions at ITU-T portal is 18th June 2024 and therefore NWG-15 has to finalise the contributions by 13th June 2024. Members are therefore requested to expedite the finalise their contribution immediately.

to also requested the members to attend online correspondence discussion meetings called by the concerned Rapporteurs between April to June 2024.

After the introduction, Sh. Rakesh Desai explained the members on potential work-items for contributions and also requested other members to actively participate in the discussions on various contributions.

4. Contribution on - Revision of Recommendation L.360: “Operations support system requirements for infrastructure and network elements management using Identification data (ID) technology” (Q7/15)

Sh.Sriram from M/s Aprecomm presented the draft of new contribution on ‘Network Monitoring Expert (NME)’ corresponding to the **Revision of Recommendation L.360** and explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. Sh. Desai requested the contributors to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Sh.Rakesh Desai, TEC and S/Sh. Guharajan Shivakumar & Sriram from M/s

Aprecomm.)

5. Contribution on- 'L.nis-Technical and functional requirements for network infrastructures sharing' (Q7/15)

Sh.Desai presented the draft of new contribution on a new work-item – 'L.nis-Technical and functional requirements for network infrastructures sharing' and explained the queries of the participants. He stated that India is one of the supporting members of this new work-item. He also explained that inputs from TSPs/ISPs/IPs have been sought with the help of CS Division of DOT HQ and some service providers have submitted their inputs. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Sh.Rakesh Desai, Sh. Anil Gehlot, Ms. Mudita Chandra)

6. Contribution on 'Revision of ITU-T Recommendation for L.102 "Optical fibre cables for aerial application". (Q5/15)

Ms. Mudita, ADET (Tx), TEC presented the draft of contribution on 'Revision of ITU-T Recommendation for L.102 "Optical fibre cables for aerial application" and explained the queries of the participants. She stated that main texts proposed for revision are regarding adding ADSS optical fibre cables deployed along electric power lines, creep and stringing through the sheave, etc. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Sh. Vijay Dixit, Ms. Mudita Chandra, Sh. Sudipta Bhaumik and Sh. Bikram Nayak from STL)

7. Contribution to include new Appendix to the ITU-T Rec. L.104 (ex. L.67) on "Small count optical fibre cables for indoor applications". (Q5/15)

Ms. Mudita, ADET (Tx), TEC presented the draft of contribution proposing to include new Appendix to the ITU-T Rec. L.104 on "Small count optical fibre cables for indoor applications" and explained the queries of the participants. She stated that proposed new Appendix II describes the Indian experience of small count optical fibre cables for indoor applications. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Sh.Rakesh Desai, Sh. Vijay Dixit, Ms. Mudita Chandra, Sh. Roshan Kumar from Westcoast Optilinks Ltd)

8. Contribution on- 'Discussions on Characteristics of Coordinated Multi-Point

(CoMP) transmission systems to support IMT-2030/6G and future networks by fronthauling in G-PON” (Q11/15)

Ms. Preeti Samhita Pati presented the draft of new contribution on a new work-item – ‘Characteristics of Coordinated Multi-Point (CoMP) transmission systems to support IMT-2030/6G’ and explained the queries of the participants. She highlighted the advantages of Coordinated Multi-Point (CoMP) transmission systems to enhance 6G services and also its relevance w.r.to synchronization in TWDM-PON. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Prof. Goutam Das, Preeti Samhita Pati, Sourabh Bhandarkar and Sourav Dutta from IIT Kharagpur, Sh.Rakesh Desai)

9. Contribution on revision of Baseline draft of ‘ G.fso-Terrestrial free space optics for mobile backhaul with short reach interfaces’ (Q6/15)

Sh.Desai presented the draft of contribution proposing revision of baseline draft of ‘ G.fso-Terrestrial free space optics for mobile backhaul with short reach interfaces’ and explained the queries of the participants. He stated that Indian experts have participated in correspondence discussions during Q6/15 meeting and prepared this draft as per the inputs / suggestions received during correspondence meeting. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution **by 3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Sh.Rakesh Desai, Ms. Mudita Chandra , Dr. A. Paventhan and S Kirubakaran from ERNET)

10. Contribution on ‘G.Suppl.G65x: Standardization roadmap for SDM optical fibres including Multi-Core Fibres’ (Q5/15)

Sh. Desai presented the draft of contribution submitted by IIT-Madras experts stating that several suggestions have been proposed to the working document circulated by Q5/15 on new supplement G Suppl.G.65x on ‘Standardization roadmap for optical fibres and cables for space division multiplexing’. As Multi-Core Fibres (MCF) is being studied in DOT’s ongoing project namely ‘Advance Optical Communication Test Bed Project’ at IIT-Madras, Indian experts may give their inputs / suggestions on this contribution. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution **by 3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Prof. Deepa and Mr. Sobhanbabu from IIT-Madras, Sh.Rakesh Desai, Sh.Arvind from STL)

11. Contribution on distributed fibre optic sensing (DFOS) systems (Q6/15)

Sh. Sudipta Bhaumik informed the members that contribution on DFOS is at advance stage and will be submitted in couple of days. Sh.Desai requested the contributors to submit the draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Contributors: Experts from M/s Sterlite Technologies Limited (STL) viz. S/Sh. Badri Gomatam; Nagaraju Bezawada; Sudipta Bhaumik and Sh.Rakesh Desai from TEC)

12. Contribution to include new Appendix to the ITU-T Rec. L.330 on “Telecommunication infrastructure facility management” (Q7/15)

Sh.Desai stated that Indian experts have participated correspondence discussions during Q7/15 meeting on revision of **ITU-T Rec. L.330** and proposed the Q6/15 to highlight the Indian experience on GRP (glass reinforced plastic) poles. He stated that draft is at advance stage for Indian experience on GRP (glass reinforced plastic) poles. The contributors were requested to submit the draft contribution by **3 p.m. 12th June 2024** for further discussions in the next meeting of NWG-15 scheduled on **13th June 2024 at 12 hrs.**

(Action by: Sh.Rakesh Desai, Sh. Bikram Mohanty from M/s Custom Wraps India Pvt Ltd)

13. Contributions to APT for the forthcoming WTSA-24:-

Sh. Desai, while referring to earlier discussions, informed the group that India has proposed revision in WTSA Resolution no.40- Regulatory & policy aspects of the work of the ITU-T which was discussed in last two meeting of APT. Experts from Iran has suggested some modifications to the Indian proposal and final discussions will be held in the forthcoming meeting of APT scheduled in the last week of April 2024. NWG-15 members were requested to give their inputs in this regard by **12th June 2024.**

(Action by: All members)

14. Sh. Rakesh Desai once again requested all the NWG-15 members to obtain TIES account by registering on ITU portal and also register for the forthcoming SG-15 & WTSA-24 meeting.

15. The key action points emerged from the meeting are as under: -

- i. Members / experts are requested to submit contributions as per the timeline as discussed above.
- ii. Members may examine the WTSA resolution no.24 and propose amendments if any by 12th June 2024..
- iii. **Next (11th) meeting of NWG-15 will be held at 12 hrs on 13th June 2024. Meeting link will remain same i.e <https://cdotmeet.cdote.in/vmeet/raj-16i-e8j-oxq>**

16. The meeting ended with the vote of thanks to the chair.

(Rajesh Tripathi)

Convener, NWG-15 & Director (FA), TEC

To,

All the participants / Members of the NWG-15.

Copy to: Sr.DDG & Head, TEC for kind information.

List of Participants

1. Sh. Rakesh Desai DDG (FA), TEC
2. Sh. Rajesh Tripathi, Dir(FA), TEC
3. Ms. Jyoti Roat, TEC
4. Ms. Mudita Chandra, TEC
5. Sh. Abhijit Mitra, IIIT Delhi
6. Sh. Anil Kr Ranjan, DDG Security UP East LSA.
7. Sh. Anil Kumar Gehlot, DDG(CS) DoT
8. Sh. Arvind Mishra, M/s STL
9. Sh. Bikram Kesharee Nayak, M/s Sterlite Technologies Limited
10. Sh. Goutam Das, IIT Kharagpur
11. Sh. Himmat Lal, M/s C- DOT
12. Sh. M K Gupta, DDG (USOF), DoT
13. Sh. Nagaraju Bezawada, M/s Sterlite Technologies Limited,
14. Sh. P.P Sonkusare, M/s Veekay
15. Ms. Preeti Samhita Pati, IIT Kharagpur
16. Sh. Rajeev Chaturvedi, M/s Dura-Line India Pvt Ltd
17. Sh. Rajeev Kumar Kaushik, BSNL
18. Sh. Ram Chet, M/s C-DOT
19. Sh. S Kirubakaran, M/s ERNET India
20. Sh. Sobhan, M/s IIT-Madras
21. Sh. Sourabh Bhandarkar, Indian Institute of Technology Kharagpur
22. Sh. Sourav Dutta, IIT Kharagpur
23. Sh. Sriram N, M/s Aprecomm Pvt Ltd
24. Sh. Sudipta Bhaumik, M/s Sterlite Technologies Ltd
25. Sh. Yogesh, M/s C-DOT
26. Ms. Yashita Mudgal, TEC
27. Sh. Sourav Roy, RA(Tx), TEC
28. Sh. Akash Arya, TEC
29. Ms. Joja J M, TEC
